NOWCASTING · LECTURE

Real-Time Economic Nowcasting

State space models, mixed-frequency data, the Kalman filter and EM

Concept · State space form · Kalman filter · Hand-calculation examples · EM algorithm · Real-time process

Source paper: Lee, H., Choi, D., & Kim, Y. (2022). Development of a current-quarter economic outlook (GDP nowcasting) system combining a dynamic factor model and deep learning algorithms.

01

PART 1

Concept and Model

Why nowcast, the state space form, mixed-frequency data and the Kalman filter

Nowcasting estimates the present, not the future

1. The concept of nowcasting — “Now” + “Forecasting”: estimating today's economic state in real time

1.1 Why it is needed

- Key indicators such as GDP have a publication lag
- Quarterly GDP is released about 1–2 months after the quarter ends
- Policy and investment decisions need the current state of the economy
- High-frequency data (monthly, weekly, daily) are used to estimate the low-frequency (quarterly) indicator

1.2 Core idea

- Current-quarter GDP growth is observed in the last month of the quarter
- Values for the earlier months are filled in by filtering / smoothing
- This yields the nowcast for the current quarter

Two equations link what we see to what we infer

2.1 Basic structure — a state space model consists of two equations

Observation equation

$$y_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H_t)$$

State equation

$$\alpha_{t+1} = T_t \alpha_t + R_t \eta_t, \quad \eta_t \sim N(0, Q_t)$$

Where

y_t:	Observable variable (e.g., a monthly indicator)
α_t:	Latent state (e.g., quarterly GDP growth)
Z_t, T_t, R_t:	System matrices
ε_t, η_t:	Observation error, state error

Quarterly GDP is a weighted sum of monthly values

3. Mixed-frequency data — 3.1 quarter–month conversion · 3.2 state space form

Quarterly GDP vs. monthly indicators

$$y_t^* = \omega_1 x_{t,1} + \omega_2 x_{t,2} + \omega_3 x_{t,3}$$

Linear-combination weights

$$\omega_1 + \omega_2 + \omega_3 = 1$$

End of quarter

$$(m = 3)\colon\ y_t = [1 \quad 0 \quad 0 \quad 0]\alpha_t + \varepsilon_t$$

Mid-quarter: monthly data only

$$(m = 1, 2)\ :\ \ y_t = \text{missing}$$

3.2 State vector

$$\alpha_t = \begin{bmatrix} y_t^* \\ x_{t,1} \\ x_{t,2} \\ x_{t,3} \end{bmatrix}$$

Each month the filter predicts, then updates

4. Real-time estimation with the Kalman filter — 4.1 the algorithm

Prediction step

$$\alpha_{t|t-1} = T_t \alpha_{t-1|t-1}$$

$$P_{t|t-1} = T_t P_{t-1|t-1} T_t' + R_t Q_t R_t'$$

- v_t: Prediction error (innovation)
- F_t: Prediction-error variance
- K_t: Kalman gain

Update step

$$v_t = y_t - Z_t \alpha_{t|t-1}$$

$$F_t = Z_t P_{t|t-1} Z_t' + H_t$$

$$K_t = P_{t|t-1} Z_t' F_t^{-1}$$

$$\alpha_{t|t} = \alpha_{t|t-1} + K_t v_t$$

$$P_{t|t} = P_{t|t-1} - K_t F_t K_t'$$

02

PART 2

Kalman Filter by Hand

Example 1 AR(1) · Example 2 quarterly–monthly mixed frequency · Example 3 multivariate system

Example 1 treats GDP growth as a noisy AR(1)

5. Hand-calculation example 1: a simple AR(1) model — 5.1 model setup

Assume GDP growth follows an AR(1) process

$$y_t = \phi y_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2)$$

It is observed with noise

$$z_t = y_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

State space form

State equation

$$y_t = \phi y_{t-1} + \eta_t$$

Observation equation

$$z_t = y_t + \varepsilon_t$$

Parameter values

$$\phi = 0.8,\ \sigma_\eta^2 = 1,\ \sigma_\varepsilon^2 = 0.5$$

One update turns observation 1.5 into 1.230

5.2 Initial condition (t = 0) and 5.3 the t = 1 nowcast — prediction, observation, update

5.2 Initial condition

$$y_{0|0} = 0, \quad P_{0|0} = 2$$

Prediction step

$$y_{1|0} = \phi y_{0|0} = 0.8 \times 0 = 0$$

$$P_{1|0} = \phi^2 P_{0|0} + \sigma_\eta^2 = 0.64 \times 2 + 1 = 2.28$$

Observation

$$z_1 = 1.5$$

Update step

$$v_1 = z_1 - y_{1|0} = 1.5 - 0 = 1.5$$

$$F_1 = P_{1|0} + \sigma_\varepsilon^2 = 2.28 + 0.5 = 2.78$$

$$K_1 = \frac{P_{1|0}}{F_1} = \frac{2.28}{2.78} = 0.820$$

$$y_{1|1} = y_{1|0} + K_1 v_1 = 0 + 0.820 \times 1.5 = 1.230$$

$$P_{1|1} = P_{1|0} - K_1 F_1 K_1 = 2.28 - 0.820 \times 2.78 \times 0.820 = 0.410$$

Result: the GDP growth nowcast at t = 1 is 1.230

Example 2 averages three monthly AR(1) values

6. Hand-calculation example 2: quarterly–monthly mixed frequency — 6.1 model setup

Quarterly GDP is the three-month average

$$y_t^* = \frac{1}{3}(x_{t,1} + x_{t,2} + x_{t,3})$$

Monthly indicator is AR(1)

$$x_{t,m} = 0.7x_{t,m-1} + \eta_{t,m}, \quad \eta_{t,m} \sim N(0,0.8)$$

Observation error

$$\sigma_\varepsilon^2 = 0.3$$

January's reading lifts the estimate to 2.049

6.2 Q1, January (t = 1, m = 1) — initial condition, prediction, observation, update

Initial condition

$$x_{0,3|0,3} = 2.0, \quad P_{0,3|0,3} = 1.0$$

Prediction (t = 1, m = 1)

$$x_{1,1|0,3} = 0.7 \times 2.0 = 1.4$$

$$P_{1,1|0,3} = 0.7^2 \times 1.0 + 0.8 = 0.49 + 0.8 = 1.29$$

Observation

$$z_{1,1} = 2.2$$

Update

$$v_{1,1} = 2.2 - 1.4 = 0.8$$

$$F_{1,1} = 1.29 + 0.3 = 1.59$$

$$K_{1,1} = \frac{1.29}{1.59} = 0.811$$

$$x_{1,1|1,1} = 1.4 + 0.811 \times 0.8 = 2.049$$

$$P_{1,1|1,1} = 1.29 - 0.811 \times 1.59 \times 0.811 = 0.243$$

With January alone the GDP nowcast is 1.496

6.3 GDP nowcast at end-January (interim estimate) — only the January indicator is observed

Method 1: use simple predictions

$$x_{1,1|1,1} = 2.049 \text{ (observed)}$$

$$x_{1,2|1,1} = 0.7 \times 2.049 = 1.434 \text{ (forecast)}$$

$$x_{1,3|1,1} = 0.7 \times 1.434 = 1.004 \text{ (forecast)}$$

$$y^*_{1|\text{Jan}} = \frac{1}{3}(2.049 + 1.434 + 1.004) = \frac{4.487}{3} = 1.496$$

End-January GDP nowcast: 1.496 (high uncertainty)

February's reading pulls the estimate to 1.710

6.4 Q1, February (t = 1, m = 2) — prediction, observation, update

Prediction

$$x_{1,2|1,1} = 0.7 \times 2.049 = 1.434$$

$$P_{1,2|1,1} = 0.7^2 \times 0.243 + 0.8 = 0.919$$

Observation

$$z_{1,2} = 1.8$$

Update

$$v_{1,2} = 1.8 - 1.434 = 0.366$$

$$F_{1,2} = 0.919 + 0.3 = 1.219$$

$$K_{1,2} = \frac{0.919}{1.219} = 0.754$$

$$x_{1,2|1,2} = 1.434 + 0.754 \times 0.366 = 1.710$$

$$P_{1,2|1,2} = 0.919 - 0.754^2 \times 1.219 = 0.226$$

Two observed months raise the nowcast to 1.669

6.5 GDP nowcast at end-February (improved estimate) — re-estimate after January–February

1. Smooth January
Feb information flows back to Jan

$$x_{1,1|1,2} = 2.100 \text{ (smoothed)}$$

2. GDP calculation

$$x_{1,1|1,2} = 2.100 \text{ (Jan, smoothed)}$$

$$x_{1,2|1,2} = 1.710 \text{ (Feb, observed)}$$

$$x_{1,3|1,2} = 0.7 \times 1.710 = 1.197 \text{ (Mar forecast)}$$

$$y^*_{1|\text{Feb}} = \tfrac{1}{3}(2.100 + 1.710 + 1.197) = \tfrac{5.007}{3} = 1.669$$

End-February GDP nowcast: 1.669 (uncertainty falls)

March's surprise moves the estimate to 2.177

6.6 Q1, March (t = 1, m = 3) — prediction, observation, update

Prediction

$$x_{1,3|1,2} = 0.7 \times 1.710 = 1.197$$

$$P_{1,3|1,2} = 0.7^2 \times 0.226 + 0.8 = 0.911$$

Observation

$$z_{1,3} = 2.5$$

Update

$$v_{1,3} = 2.5 - 1.197 = 1.303$$

$$F_{1,3} = 0.911 + 0.3 = 1.211$$

$$K_{1,3} = \frac{0.911}{1.211} = 0.752$$

$$x_{1,3|1,3} = 1.197 + 0.752 \times 1.303 = 2.177$$

$$P_{1,3|1,3} = 0.911 - 0.752^2 \times 1.211 = 0.226$$

With the full quarter the final nowcast is 2.063

6.7 GDP nowcast at end-March (final estimate) — after all information in the quarter is observed

The first two months are re-estimated with the Kalman smoother

$$y_1^* = \frac{1}{3}(x_{1,1|1,3} + x_{1,2|1,3} + x_{1,3|1,3})$$

Smoothing the whole quarter — March information propagates back to January–February

$$x_{1,1|1,3} \approx 2.132, \quad x_{1,2|1,3} \approx 1.880, \quad x_{1,3|1,3} = 2.177$$

$$y^*_{1|\text{Mar}} = \tfrac{1}{3}(2.132 + 1.880 + 2.177) = \tfrac{6.189}{3} = 2.063$$

End-March (final) GDP nowcast: 2.063 (lowest uncertainty)

More information moves the estimate, cuts doubt

6.8 Summary of real-time updates

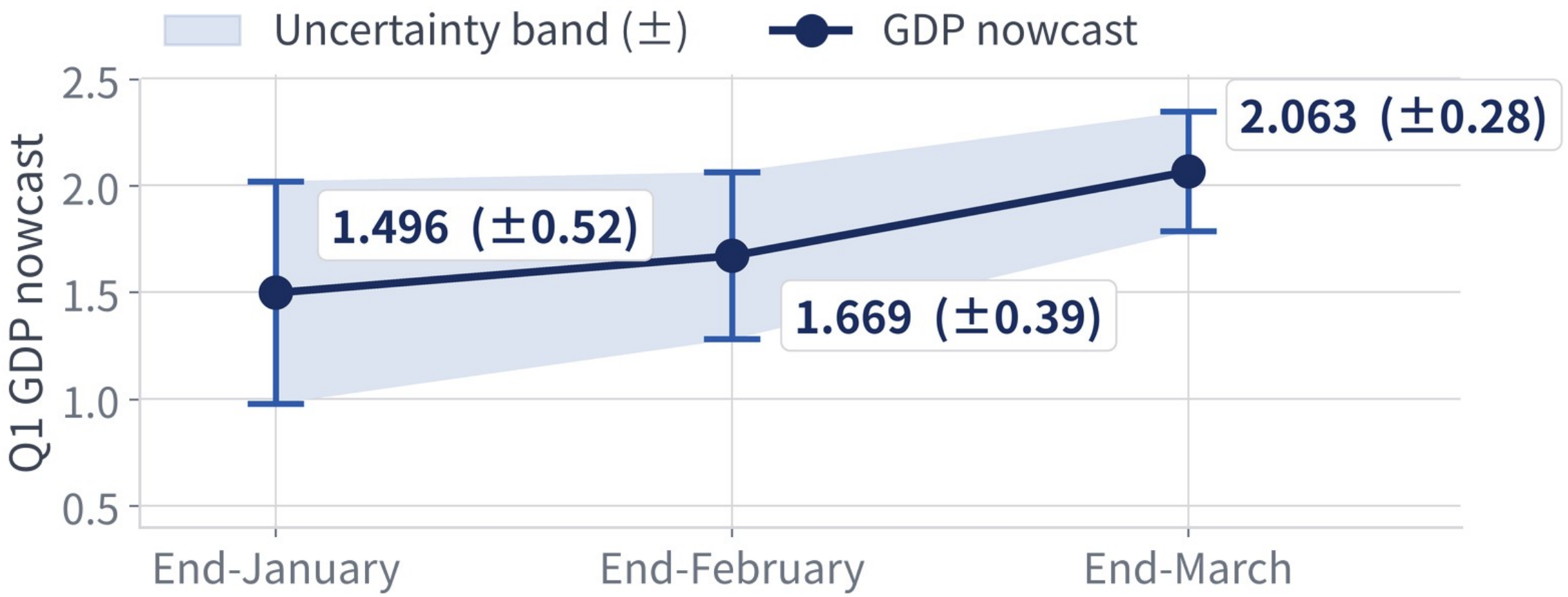
Time	Information observed	GDP nowcast	Uncertainty
End-January	$x_{1,1}$ only	1.496	High (±0.52)
End-February	$x_{1,1}, x_{1,2}$	1.669	Medium (±0.39)
End-March	$x_{1,1}, x_{1,2}, x_{1,3}$	2.063	Low (±0.28)

Interpretation

- As information accumulates the GDP estimate changes — and its uncertainty falls at the same time. This is the heart of real-time nowcasting.

The nowcast rises as its uncertainty narrows

Q1 GDP nowcast at each month-end with its uncertainty (±) · values from the worked example



Example 3 builds GDP from two monthly indicators

7. Hand-calculation example 3: a multivariate system — 7.1 model setup

Quarterly GDP from two indicators

$$y_t^* = 0.6x_t^{(1)} + 0.4x_t^{(2)}$$

State vector

$$\alpha_t = \begin{bmatrix} y_t^* \\ x_t^{(1)} \\ x_t^{(2)} \end{bmatrix}$$

State equation

$$\begin{bmatrix} y_{t+1}^* \\ x_{t+1}^{(1)} \\ x_{t+1}^{(2)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 0.6 \end{bmatrix} \begin{bmatrix} y_t^* \\ x_t^{(1)} \\ x_t^{(2)} \end{bmatrix} + \begin{bmatrix} \eta_t^y \\ \eta_t^{(1)} \\ \eta_t^{(2)} \end{bmatrix}$$

Covariance

$$Q = \mathrm{diag}(0, 0.5, 0.4)$$

Observation: indicator 1 only

$$z_t = [0 \quad 1 \quad 0]\alpha_t + \varepsilon_t, \quad \sigma_\varepsilon^2 = 0.2$$

The prediction step inflates the prior covariance

7.2 Initial condition (t = 0) and 7.3 prediction at t = 1

7.2 Initial condition

$$\alpha_{0|0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad P_{0|0} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7.3 Prediction at t = 1

$$\alpha_{1|0} = T\alpha_{0|0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Prior covariance

$$P_{1|0} = TP_{0|0}T' + Q = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.64 + 0.5 & 0 \\ 0 & 0 & 0.36 + 0.4 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1.14 & 0 \\ 0 & 0 & 0.76 \end{bmatrix}$$

Observing indicator 1 alone yields a 1.532 nowcast

7.4 Observation and update at t = 1 — only indicator 1 is observed

$z_1 = 3.0$ (indicator 1 only)

$$Z = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

$$v_1 = z_1 - Z\alpha_{1|0} = 3.0 - 0 = 3.0$$

$$F_1 = ZP_{1|0}Z' + \sigma_\varepsilon^2 = 1.14 + 0.2 = 1.34$$

Kalman gain and state update

$$K_1 = P_{1|0}Z'F_1^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1.14 & 0 \\ 0 & 0 & 0.76 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{1}{1.34} = \begin{bmatrix} 0 \\ 0.851 \\ 0 \end{bmatrix}$$

$$\alpha_{1|1} = \alpha_{1|0} + K_1 v_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.851 \\ 0 \end{bmatrix} \times 3.0 = \begin{bmatrix} 0 \\ 2.553 \\ 0 \end{bmatrix}$$

GDP nowcast

$$y_1^* = 0.6 \times 2.553 + 0.4 \times 0 = 1.532$$

Result: 1.532

03

PART 3

EM Algorithm

Parameter estimation and missing-data handling — theory, Shumway–Stoffer, and a numeric example

Nowcasting must estimate the parameters too

8.1 Problem setting and motivation — the premise of the Kalman filter

The filter assumes θ is known

$$\theta = (\phi, \sigma_\eta^2, \sigma_\varepsilon^2, \ldots)$$

In practice the parameters must be estimated from data

EM solves the missing values and the parameters at the same time

The double problem of nowcasting

1. Missing values: indicators are released at different times (ragged edge)
2. Unknown parameters: the parameters of the state space model are not known
3. EM combines naturally with the Kalman smoother

EM breaks the chicken-and-egg loop by alternating

8.2.1 The “chicken or egg” dilemma — EM's solution: alternate the two steps repeatedly

The dilemma

- Know the parameters → can estimate the missing values (Kalman filter)
- Know the missing values → can estimate the parameters (MLE)

EM loop

1. Initial value $\theta^{(0)}$ (e.g., $\varphi = 0.5$)
2. E-step: fix θ, compute expected missing values with the Kalman smoother
3. M-step: estimate θ by MLE from the filled-in data
4. Convergence check — YES → stop / NO → repeat the E-step

Convergence check $|\theta^{(k+1)} - \theta^{(k)}| < \epsilon$?

DEFINITION

EM maximizes the likelihood of the observed data

8.2.2 Mathematical definition — goal and the idea of the EM algorithm

Goal: find θ that maximizes the log-likelihood using only the observed data

$$\max_{\theta} \log p(Y_{\text{obs}}|\theta)$$

But part of the data is hidden

- Latent variables / missing values — call them α

EM idea

- Predict the hidden α with the current θ (E-step), then update θ with that prediction (M-step)

The E-step estimates α; only the M-step moves θ

8.2.2 Mathematical definition — E-step (expectation) and M-step (maximization)

E-step (Expectation step)

Expected log-likelihood of complete data

$$Q(\theta \mid \theta^{(k)}) = E\left[\log p(Y_{\text{obs}}, \alpha \mid \theta) \mid Y_{\text{obs}}, \theta^{(k)}\right]$$

M-step (Maximization step)

Choose the θ that maximizes Q

$$\theta^{(k+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(k)})$$

This step estimates α; it does not update θ

Only here is θ updated

Each iteration never lowers the likelihood

8.2.2 Mathematical definition — iteration, guaranteed convergence, and what to remember

Iteration

$$\theta^{(k)} \rightarrow \text{E-step} \rightarrow \text{M-step} \rightarrow \theta^{(k+1)} \rightarrow \cdots$$

Guaranteed convergence

- Log-likelihood rises every iteration

Key takeaway

Step	What it estimates	Input	Output
E-step	Hidden data α	$\theta^{(k)}$	$Q(\theta \mid \theta^{(k)})$
M-step	Parameters θ	Q	$\theta^{(k+1)}$

EM for state space models runs on the Kalman filter

8.3 Link to the Kalman filter: the Shumway–Stoffer algorithm — 8.3.1 the state space model again

Observation equation

$$y_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H_t)$$

State equation

$$\alpha_{t+1} = T_t \alpha_t + \eta_t, \quad \eta_t \sim N(0, Q_t)$$

Parameters to estimate (or their elements)

$$\theta = (T, Q, Z, H)$$

E-step: forward filter, then backward smoother

8.3.2 E-step = Kalman smoother — the E-step needs the expected sufficient statistics

Forward: Kalman filter

$$\alpha_{t|t}, \quad P_{t|t} \quad (t = 1, \ldots, T)$$

Forward pass

- Exactly as in Section 4
- Missing values handled automatically (that observation is skipped)

Backward: Kalman smoother

$$\alpha_{t|T} = \alpha_{t|t} + J_t(\alpha_{t+1|T} - \alpha_{t+1|t})$$

$$P_{t|T} = P_{t|t} + J_t(P_{t+1|T} - P_{t+1|t})J_t'$$

where

$$J_t = P_{t|t}T'P_{t+1|t}^{-1}$$

Three smoothed sums feed the M-step

8.3.2 E-step — lag-one covariance smoother (covariance between time points) and sufficient statistics

Lag-one covariance smoother

$$P_{t,t-1|T} = P_{t|t}J'_{t-1} + J_t(P_{t+1,t|T} - TP_{t|t})J'_{t-1}$$

Sufficient statistics

$$S_1 = \sum_{t=1}^{T}(\alpha_{t|T}\alpha'_{t|T} + P_{t|T})$$

$$S_2 = \sum_{t=1}^{T-1}(\alpha_{t+1|T}\alpha'_{t|T} + P_{t+1,t|T})$$

$$S_3 = \sum_{t=1}^{T-1}(\alpha_{t|T}\alpha'_{t|T} + P_{t|T})$$

The M-step updates T, Q, Z and H in closed form

8.3.3 M-step = parameter update

Transition matrix T

$$T^{(k+1)} = S_2(S_3)^{-1}$$

Observation-equation parameters, similarly

$$Z^{(k+1)}, \quad H^{(k+1)}$$

State covariance Q

$$Q^{(k+1)} = \frac{1}{T-1}\sum_{t=1}^{T-1}[(\alpha_{t+1|T} - T^{(k+1)}\alpha_{t|T})(\alpha_{t+1|T} - T^{(k+1)}\alpha_{t|T})' + P_{t+1|T} + T^{(k+1)}P_{t|T}(T^{(k+1)})' - T^{(k+1)}P_{t,t+1|T} - P_{t+1,t|T}(T^{(k+1)})']$$

In short

$$Q^{(k+1)} = \frac{1}{T-1}[S_1^{(2)} - T^{(k+1)}S_2' - S_2(T^{(k+1)})' + T^{(k+1)}S_3(T^{(k+1)})']$$

A three-point AR(1) with one gap tests EM by hand

8.4 Hand-calculation example: AR(1) + missing value + parameter estimation — 8.4.1 model setup

Model

$$y_t = \phi y_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2)$$

$$z_t = y_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

Parameters to estimate

$$\theta = (\phi, \sigma_\eta^2, \sigma_\varepsilon^2)$$

Initial values

$$\phi^{(0)} = 0.5,\ \sigma_\eta^{(0)2} = 1.0,\ \sigma_\varepsilon^{(0)2} = 0.5$$

Data

$$z_1 = 1.0 \text{ (observed)}$$

$$z_2 = \text{missing}$$

$$z_3 = 2.0 \text{ (observed)}$$

At t = 1 the filter updates on the observed 1.0

8.4.2 Iteration 1: E-step (Kalman filter) — initial condition and t = 1 (observed)

Initial condition

$$y_{0|0} = 0, \quad P_{0|0} = 2.0$$

t = 1 (observed) — prediction

$$y_{1|0} = 0.5 \times 0 = 0$$

$$P_{1|0} = 0.5^2 \times 2.0 + 1.0 = 1.5$$

Update

$$v_1 = 1.0 - 0 = 1.0$$

$$F_1 = 1.5 + 0.5 = 2.0$$

$$K_1 = 1.5/2.0 = 0.75$$

$$y_{1|1} = 0 + 0.75 \times 1.0 = 0.75$$

$$P_{1|1} = 1.5 - 0.75 \times 2.0 \times 0.75 = 0.375$$

At t = 2 the value is missing, so prediction stands

8.4.2 Iteration 1: E-step (Kalman filter) — t = 2 (missing)

Prediction

$$y_{2|1} = 0.5 \times 0.75 = 0.375$$

$$P_{2|1} = 0.5^2 \times 0.375 + 1.0 = 1.094$$

Update: missing, so the prediction is kept

$$y_{2|2} = y_{2|1} = 0.375$$

$$P_{2|2} = P_{2|1} = 1.094$$

At t = 3 the observation 2.0 lifts the state to 1.489

8.4.2 Iteration 1: E-step (Kalman filter) — t = 3 (observed)

Prediction

$$y_{3|2} = 0.5 \times 0.375 = 0.188$$

$$P_{3|2} = 0.5^2 \times 1.094 + 1.0 = 1.273$$

Update

$$v_3 = 2.0 - 0.188 = 1.812$$

$$F_3 = 1.273 + 0.5 = 1.773$$

$$K_3 = 1.273/1.773 = 0.718$$

$$y_{3|3} = 0.188 + 0.718 \times 1.812 = 1.489$$

$$P_{3|3} = 1.273 - 0.718 \times 1.773 \times 0.718 = 0.358$$

Smoothing fills the gap at t = 2 with 0.933

8.4.3 Iteration 1: E-step (Kalman smoother) — t = 2 smoothing (3 → 2 backward)

t = 2 smoothing (3 → 2 backward)

$$J_2 = P_{2|2}\phi(P_{3|2})^{-1} = 1.094 \times 0.5/1.273 = 0.429$$

$$y_{2|3} = y_{2|2} + J_2(y_{3|3} - y_{3|2})$$

$$y_{2|3} = 0.375 + 0.429 \times (1.489 - 0.188) = 0.933$$

$$P_{2|3} = P_{2|2} + J_2(P_{3|3} - P_{3|2})J_2'$$

$$P_{2|3} = 1.094 + 0.429 \times (0.358 - 1.273) \times 0.429 = 0.925$$

Smoothing back to t = 1 revises it to 0.845

8.4.3 Iteration 1: E-step (Kalman smoother) — t = 1 smoothing (2 → 1 backward)

t = 1 smoothing (2 → 1 backward)

$$J_1 = P_{1|1}\phi(P_{2|1})^{-1} = 0.375 \times 0.5/1.094 = 0.171$$

$$y_{1|3} = y_{1|1} + J_1(y_{2|3} - y_{2|1})$$

$$y_{1|3} = 0.75 + 0.171 \times (0.933 - 0.375) = 0.845$$

$$P_{1|3} = 0.375 + 0.171 \times (0.925 - 1.094) \times 0.171 = 0.370$$

The smoothed series uses all three periods

8.4.3 Iteration 1: E-step (Kalman smoother) — smoothed results (whole series)

Smoothed results (whole series)

$$y_{1|3} = 0.845 \text{ (Jan)}$$

$$y_{2|3} = 0.933 \text{ (Feb, missing-value estimate)}$$

$$y_{3|3} = 1.489 \text{ (Mar)}$$

Reading the result

- The missing February value is now an estimate
- Every point uses past and future observations

The M-step raises φ from 0.5 to 0.865

8.4.4 Iteration 1: M-step (parameter update) — sufficient statistics and the φ update

Sufficient statistics

$$S_1 = y_{1|3}^2 + y_{2|3}^2 + y_{3|3}^2 + P_{1|3} + P_{2|3} + P_{3|3}$$

$$S_1 = 0.845^2 + 0.933^2 + 1.489^2 + 0.370 + 0.925 + 0.358 = 5.455$$

$$S_2 = y_{2|3}y_{1|3} + y_{3|3}y_{2|3} + P_{2,1|3} + P_{3,2|3}$$

$$S_2 = 0.933 \times 0.845 + 1.489 \times 0.933 + 0.159 + 0.154 = 2.491$$

$$S_3 = y_{1|3}^2 + y_{2|3}^2 + P_{1|3} + P_{2|3}$$

$$S_3 = 0.845^2 + 0.933^2 + 0.370 + 0.925 = 2.880$$

φ update

$$\phi^{(1)} = \frac{S_2}{S_3} = \frac{2.491}{2.880} = 0.865$$

Up from 0.5 to 0.865 — the data suggest higher autocorrelation

The M-step raises the state noise from 1.0 to 1.108

8.4.4 Iteration 1: M-step — state-noise variance update

State-noise variance update

$$\sigma_\eta^{(1)2} = \frac{1}{2}\sum_{t=1}^{2}\left[\left(y_{t+1|3} - \phi^{(1)} y_{t|3}\right)^2 + P_{t+1|3} + \left(\phi^{(1)}\right)^2 P_{t|3} - 2\phi^{(1)}\, P_{t+1,t|3}\right]$$

Calculation

$$= \tfrac{1}{2}[(0.933 - 0.865 \times 0.845)^2 + 0.925 + 0.865^2 \times 0.370 - 2 \times 0.865 \times 0.159 + (1.489 - 0.865 \times 0.933)^2 + 0.358 + 0.865^2 \times 0.925 - 2 \times 0.865 \times 0.154]$$

$$\approx 1.108$$

Observation noise uses observed points only: 0.507

8.4.4 Iteration 1: M-step — observation-noise variance update

Observation-noise variance (observed time points only)

$$\sigma_\varepsilon^{(1)2} = \frac{1}{2}\left[(z_1 - y_{1|3})^2 + (z_3 - y_{3|3})^2 + P_{1|3} + P_{3|3}\right]$$

$$= \frac{1}{2}\left[(1.0 - 0.845)^2 + (2.0 - 1.489)^2 + 0.370 + 0.358\right]$$

$$= \frac{1}{2}\left[0.024 + 0.261 + 0.728\right] = 0.507$$

One EM pass moves all three parameters

8.4.4 Iteration 1 — update summary

Parameter	Initial value	After iteration 1	Change
φ	0.500	0.865	↑ +0.365
$\sigma^2\eta$	1.000	1.108	↑ +0.108
$\sigma^2\varepsilon$	0.500	0.507	≈ +0.007

The largest move is in φ — the autocorrelation the filled-in data reveal

Repeating E and M drives φ past 1, toward 1.414

8.4.5 Iterations 2, 3, ...

E-step again with the new parameters

$\phi^{(1)} = 0.865$ — re-run the Kalman filter/smoother

The missing-value estimate changes:

$y_{2|3} = 0.933 \rightarrow y'_{2|3} \approx 1.356$

M-step again

Re-estimate parameters from the updated sufficient statistics

$\phi^{(2)} \approx 1.021$

After 10 iterations (not yet converged)

$$\phi^{(10)} = 1.328, \quad \sigma_\eta^{(10)2} = 0.117, \quad \sigma_\varepsilon^{(10)2} = 0.072$$

$$y_2^{(10)} = 1.448 \quad (\text{missing value estimate})$$

The filter runs live; EM learns the parameters

8.5.1 Comparison table — Kalman filter vs. EM algorithm

Aspect	Kalman filter	EM algorithm
Purpose	Estimate the state	Estimate parameters + state
Assumption	Parameters known	Parameters unknown
Input	θ (fixed)	Data only
Output	$\alpha_{t\|T}$	$\theta^*,\ \alpha^*_{t\|T}$
Missing values	Handled automatically	Handled + parameters improved
Computation	One pass (forward + backward)	Iterative (until convergence)
Real time	Possible (filter only)	Not possible (needs full data)

Learn offline with EM, run online with the filter

8.5.2 Practical workflow — three phases

Phase 1: Learning (historical data)

Historical data (with missing values) → run EM (10–100 iterations) → obtain the estimate θ*

Phase 2: Operation (real-time nowcasting)

Use the fixed θ* → new data arrive in real time → run only the Kalman filter (fast, one pass) → output the GDP nowcast

Phase 3: Periodic update (e.g., yearly)

Re-run EM on the accumulated new data → update θ* → return to Phase 2

EM is for offline learning (slow); the Kalman filter is for online operation (real time)

EM handles gaps and comes with guarantees

8.6 Pros, cons and practical considerations of EM — 8.6.1 advantages

1. Automatic handling of missing data

- Replaced by expected values instead of filling them explicitly
- Handles the ragged-edge problem naturally

2. Theoretical guarantees

- Monotone increase of the log-likelihood
- Convergence to a local optimum

3. Natural fit with the Kalman smoother

- E-step = Kalman smoother (an algorithm we already know)
- Minimal extra development

4. Propagation of uncertainty

- The uncertainty of the missing values (smoothed variance) is reflected in parameter estimation

CONS

EM can be slow and gives only point estimates

8.6 Pros, cons and practical considerations of EM — 8.6.2 disadvantages

1. Local optima

- May converge to different solutions depending on initial values
- Run from several starting values

2. Computational cost

- A Kalman smoother (full sample) every iteration
- Slow for large systems

3. Standard errors are hard

- EM gives only point estimates
- Hessian or bootstrap needed

4. Speed of convergence

- Sometimes very slow (100+ iterations)
- Slower the more data are missing

Good starting values come from simple models

8.6 Pros, cons and practical considerations of EM — 8.6.3 initial-value strategy

1. OLS on partial data

- Regress on the part without missing values

2. Assume a simple model

- E.g., random walk ($\varphi = 1$)
- Initialize the noise variances with the sample variance

3. Typical values from the literature

- Estimates from similar models

4. Grid search

- Try several initial values and pick the best

Stop when parameters or likelihood stop moving

8.6 Pros, cons and practical considerations of EM — 8.6.4 convergence criteria

Criterion 1: change in parameters

$$\max_i \left|\theta_i^{(k+1)} - \theta_i^{(k)}\right| < \epsilon$$

Criterion 2: change in log-likelihood

$$\left|L(\theta^{(k+1)}) - L(\theta^{(k)})\right| < \epsilon \cdot \left|L(\theta^{(k)})\right|$$

In practice

$$\epsilon = 10^{-4} \sim 10^{-6}$$

Three exercises extend the worked EM example

8.7 Exercises

Problem 1: Complete iteration 2 of Example 8.4.

Hint: re-run the Kalman filter/smoother with

$\phi^{(1)} = 0.865$

Problem 2: Run EM on the following data.

Data:

$z_1 = 2.0, z_2 = \text{missing}, z_3 = \text{missing}, z_4 = 3.5$

Model: AR(1) · initial value:

$\phi^{(0)} = 0.6$

Problem 3

- Why does the log-likelihood always increase in EM? Sketch a proof.
- Hint: Jensen's inequality

04

PART 4

The Real-Time Nowcasting Process

Information flow, state vectors, uncertainty, value of information, practice and summary

Each month's release triggers one nowcast update

9.1 Information flow — release on the 15th, nowcast at month-end

January

Jan 1 → Jan 15 (release) → Jan 31

Prior estimate (past information) → x_1 observed and updated, x_2 and x_3 predicted

End-Jan GDP nowcast: 1.496

February

Feb 1 → Feb 15 (release) → Feb 28

January information kept → x_2 observed and updated, x_3 predicted, x_1 smoothed (backward)

End-Feb GDP nowcast: 1.669

March

Mar 1 → Mar 15 (release) → Mar 31

January–February information kept → x_3 observed and updated, full smoothing

End-Mar GDP nowcast: 2.063

Each month one entry turns from forecast to data

9.2 The state vector at each point in time

End-Jan.

$$\alpha_{1,1} = \begin{bmatrix} y_1^* \\ x_{1,1|1,1} \\ x_{1,2|1,1} \\ x_{1,3|1,1} \end{bmatrix} = \begin{bmatrix} 1.496 \\ 2.049 \text{ (observed)} \\ 1.434 \text{ (forecast)} \\ 1.004 \text{ (forecast)} \end{bmatrix}$$

End-Feb.

$$\alpha_{1,2} = \begin{bmatrix} 1.669 \\ 2.100 \text{ (smoothed)} \\ 1.710 \text{ (observed)} \\ 1.197 \text{ (forecast)} \end{bmatrix}$$

End-Mar.

$$\alpha_{1,3} = \begin{bmatrix} 2.063 \\ 2.132 \text{ (smoothed)} \\ 1.880 \text{ (smoothed)} \\ 2.177 \text{ (observed)} \end{bmatrix}$$

The 95% interval narrows from ±1.02 to ±0.55

9.3 Change in uncertainty — standard deviation of the GDP estimate

End-January

$$\sigma_{GDP|\text{Jan}} = \sqrt{\frac{1}{9}(P_{1,1|1,1} + P_{1,2|1,1} + P_{1,3|1,1})}$$

$$\approx \sqrt{\tfrac{1}{9}(0.243 + 0.919 + 1.250)} \approx 0.52$$

End-February

$$\sigma_{GDP|\text{Feb}} \approx \sqrt{\tfrac{1}{9}(0.243 + 0.226 + 0.911)} \approx 0.39$$

End-March

$$\sigma_{GDP|\text{Mar}} \approx \sqrt{\tfrac{1}{9}(0.243 + 0.226 + 0.226)} \approx 0.28$$

95% confidence interval

End-January (very wide)

$$1.496 \pm 1.02 \rightarrow [0.48, 2.52]$$

End-February (medium)

$$1.669 \pm 0.76 \rightarrow [0.91, 2.43]$$

End-March (narrower)

$$2.063 \pm 0.55 \rightarrow [1.51, 2.61]$$

March's surprise was worth more than February's

9.4 Value-of-information analysis — contribution of each observation

Observation	Change in GDP	Reduction in uncertainty	Information value
January → February	+0.173	−0.13	Medium
February → March	+0.394	−0.11	High

Interpretation

- The March observation revised GDP sharply upward (2.5 is far above the forecast of 1.197)
- As information accumulates, estimation accuracy improves

Ragged edges, model choice and evaluation matter

10. Practical considerations — 10.1 data imbalance · 10.2 model selection · 10.3 evaluation

Data imbalance

- Ragged-edge problem: each indicator is released at a different time
- Some indicators observed, others missing
- The Kalman filter handles missing data naturally

Model selection

- Factor model: summarize many indicators by a few common factors
- Dynamic factor model: adds time-series dynamics
- MIDAS (Mixed Data Sampling): regression-based approach

Evaluation

- Real-time performance: backtest on historical data
- Forecast error: RMSE, MAE, etc.
- Update effect: improvement when new information is added

Three papers anchor the nowcasting literature

11. Key references

Key references

- Banbura, M., et al. (2013). “Now-casting and the real-time data flow.” ECB Working Paper.
- Giannone, D., et al. (2008). “Nowcasting: The real-time informational content of macroeconomic data.” Journal of Monetary Economics.
- Mariano, R. S., & Murasawa, Y. (2003). “A new coincident index of business cycles based on monthly and quarterly series.” Journal of Applied Econometrics.

Nowcasting = state space model + Kalman filter

12. Summary

The essentials of nowcasting

1. Goal: estimate the current state of the economy in real time
2. Method: predict low-frequency indicators from high-frequency ones (mixed frequency)
3. Tool: state space model + Kalman filter
4. Strength: handles missing data; information is updated step by step
5. Use: central-bank policy, investment decisions, business-cycle monitoring

Role of the Kalman filter

- Updates the estimate in real time whenever new information arrives
- Also provides the estimation uncertainty (variance)
- Smoothing past data re-estimates the whole series

A

APPENDIX

Hand-Calculation Checklists

Kalman filter step check · real-time update workflow · Example 3 walk-through

Check each Kalman filter step in order

Appendix: hand-calculation checklist — Kalman filter step by step

Prediction step (rows 1–2) · Update step (rows 3–7)

- [ ] $\alpha_{t|t-1} = T_t \alpha_{t-1|t-1}$ compute
- [ ] $P_{t|t-1} = T_t P_{t-1|t-1} T_t' + R_t Q_t R_t'$ compute
- [ ] $v_t = y_t - Z_t \alpha_{t|t-1}$ (prediction error)
- [ ] $F_t = Z_t P_{t|t-1} Z_t' + H_t$ (prediction-error variance)
- [ ] $K_t = P_{t|t-1} Z_t' F_t^{-1}$ (Kalman gain)
- [ ] $\alpha_{t|t} = \alpha_{t|t-1} + K_t v_t$ (state update)
- [ ] $P_{t|t} = P_{t|t-1} - K_t F_t K_t'$ (variance update)

Cautions

- Matrix dimensions match
- Variances are always positive
- Kalman gain lies between 0 and 1 (univariate case)
- Prediction errors have mean 0

Observe, update, nowcast, then next quarter

Appendix: real-time update workflow

[t=1, m=1]

January indicator observed

Kalman filter update → compute GDP_1

end-January nowcast

[t=1, m=2]

February indicator observed

Kalman filter update → compute GDP_2

end-February nowcast

[t=1, m=3]

March indicator observed

Kalman filter update + smoother → compute GDP_3

end-March final nowcast

Start → three monthly update cycles → on to the next quarter

Only indicator 1 is seen — how do we estimate GDP?

Appendix: hand-calculation checklist (Example 3) — setting

Quarterly GDP weights two indicators; only indicator 1 is observed

$$\text{GDP} = 0.6 \times \text{Indicator } 1 + 0.4 \times \text{Indicator } 2$$

Step 1: initial state (everything uncertain)	Value	Variance
GDP	?	2
Indicator 1	?	1
Indicator 2	?	1

Trusting 85% of a 3.0 surprise gives GDP 1.532

Appendix: hand-calculation checklist (Example 3) — steps 2 and 3

Step 2: indicator 1 = 3.0 observed

- Expected: indicator 1 = 0 (prior mean)
- Observed: indicator 1 = 3.0
- Error: 3.0 (way off!)

Kalman gain = 0.851: "reflect 85% of the error"

$$\text{Indicator 1 update} : 0 + 0.851 \times 3.0 = 2.553$$

Step 3: GDP estimate

```
GDP = 0.6 × 2.553 + 0.4 × 0 = 1.532
```

Why is indicator 2 zero?

- Indicator 2 is not observed
- Indicators 1 and 2 are independent (covariance = 0)
- So indicator 1's information does not pass to indicator 2

A small noise relative to the prior means high trust

Appendix: hand-calculation checklist (Example 3) — practical meaning and gain intuition

Scenario

- Indicator 1 = industrial production index (released early)
- Indicator 2 = consumer spending (released late); GDP combines the two

Interpretation

- Production came in 3.0 above expectations
- "So GDP is high too?" Consumption is still unknown, so assume 0

$$\text{GDP} = 0.6 \times (\text{high industrial production}) + 0.4 \times 0 = \text{partial rise}$$

What K = 0.851 means

$$\text{trust} = \text{prior variance} \,/\, (\text{prior variance} + \text{observation noise})$$

$$= 1.14 \,/\, (1.14 + 0.2) = 0.85$$

The gain is high: we know little, measure well

Appendix: hand-calculation checklist (Example 3) — the intuition of the Kalman gain, continued

Prior variance 1.14 is large

"We do not know much"

Observation noise 0.2 is small

"The measurement is accurate"

Therefore

Trust the observation at 85%

$$\text{trust} = \text{prior variance} \,/\, (\text{prior variance} + \text{observation noise})$$

$$= 1.14 \,/\, (1.14 + 0.2) = 0.85$$

A zero gain means no information flows to that state

Appendix: hand-calculation checklist (Example 3) — why is indicator 2 not updated?

Kalman gain vector

```
K = [0, 0.851, 0]'
```

First 0: GDP is not observed directly

Second 0.851: indicator 1 is observed

Third 0: indicator 2 is unobserved and unrelated to indicator 1

If indicators 1 and 2 were correlated

- "If industrial production rises, consumption will rise too" (positive correlation)
- Then indicator 2 would also be updated slightly
- But this example assumes independence → no propagation

Summary

$$\text{GDP} = 0.6 \times 2.553 = 1.532 \quad (\text{Indicator 2 stays at 0 : no information})$$

The filter sets trust by variance, then weighs news

Appendix: hand-calculation checklist (Example 3) — three-line summary and key message

Three-line summary

1. Indicator 1 observed at 3.0 → 3.0 above the expectation (0)
2. Trust 85% of the error → indicator 1 updated to 2.553
3. GDP nowcast 1.532 — indicator 2 has no information and stays at 0

Key message: trust is computed from variances; news enters at the optimal ratio